

A Controlled Replication of a Self-Model Loop for Language-Model Agents: Controls, Confounds, and Finite-Horizon Path Dependence

Version 1.1 — revised after external review

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September 1, 2026

Abstract

A recently described architecture wraps a large language model (LLM) with a self-referential inference layer: a Bayesian belief over the agent’s own interaction “stance”, re-injected into generation each turn, together with a hidden deliberative monologue. The originating write-ups report strong effects but do not publish the controls needed to separate the contribution of the state *content* from the presence of an instruction, an honest task baseline, or the intrinsic inertia of the estimator. We give an independent, clean-room reimplementation of the loop’s core (stance tracker, state prefix, optional private monologue) and subject its central claims to those controls across seven small experiments on one or a few model endpoints. We find: (i) under a structured-placebo control the state’s *content* is behaviourally load-bearing only when the posterior is confident, and the private monologue is the strongest single factor—though without a matched text-length placebo we cannot isolate it as the unique causal channel; (ii) a clamped partial factorial shows that mode-labelled prefixes reliably condition reply length and question use (an instruction-following result, not a probabilistic calibration test) and that the “gate” token is a substantial, non-uniform channel rather than a minor one; (iii) on a ground-truth hidden-stance task the loop infers well above chance (18/30; $p \approx 2 \times 10^{-6}$) but shows no accuracy advantage over an honest single-shot LLM baseline, which is itself far above chance; (iv) driving the closed loop with a ramped numerical push produces substantial finite-horizon path dependence with no observed sharp jump across seven sampled levels; a state-decoupled null shows roughly two thirds of the raw separation is accumulator arithmetic, leaving an arithmetic residual of ~ 0.07 (mean vertical separation) whose size and mechanism we do not claim to have statistically identified; that residual did not collapse for dwell times up to 12 turns per level, but a 100-turn basin test shows relaxation still under way, so slow lag is not excluded; and the basin test found trajectories from opposite initial states converging and still closing at turn 100—no evidence of bistability in this limited test, not a demonstration of a single attractor. Every claim tested reproduces in a “true-but-softer” form once the missing controls are added, and several of our own stronger intermediate readings failed later controls. We make no claim about consciousness or any threshold measure; the scope is strictly the loop’s measurable behaviour. Code and data are released. This is a preprint, not a peer-reviewed result.

1 Introduction

Adding a persistent, self-referential state to an LLM agent—a belief about the agent’s own stance, updated online and fed back into generation—is an appealing idea with a growing informal liter-

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ature. A prominent recent instantiation (hereafter “the AC1 loop”, following its author’s naming) combines four ingredients: a Bayesian stance tracker; a rendered state *prefix* injected upstream of the reply; a hidden *private monologue* that deliberates before the public reply; and a second-order term modelling the interlocutor. The originating essays report large effects—for example that removing the monologue changes the output on 92–100% of measurable turns—and frame these as evidence for strong properties.

Our interest is narrower and purely methodological. Effects of this kind can arise for reasons that have nothing to do with a self-model: any prefix that contains an imperative will change behaviour; a “baseline” set at chance will make any above-chance system look impressive; and an estimator that accumulates evidence will look path-dependent whether or not the surrounding loop is—with decay, through inertia; without decay, by construction, since an undecayed accumulator never forgets. The originating write-ups do not report the controls that separate these. This paper supplies them, on an independent reimplementaion, and reports what survives.

Scope and non-claims. We study only the measurable behaviour of the loop. We make **no** claim about the Ψ -threshold, consciousness, or the wider conjecture the architecture is embedded in; those questions are neither tested nor implied here. Where a claim holds we say so; where it holds only in a softer form, or rests on an unresolved confound, we say that too.

2 The loop

The state is a Dirichlet posterior over five stance modes {exploring, evaluating, overwhelmed, asserting, disengaging}. Each turn:

1. a **sensor** (an LLM classification with confidence) reads the latest line and yields a per-mode likelihood ℓ and a scalar evidence e ;
2. the tracker updates $\alpha \leftarrow \rho\alpha + e\ell$, with a mean-step cap, an α floor, and a concentration cap;
3. the posterior $p = \alpha/\kappa$ is rendered into a **prefix** (the five values, the dominant mode, a *gate* flag, and an optional per-mode *strategy* line);
4. an optional **private monologue** is generated from the prefix and prepended, unseen, to the reply’s system prompt;
5. the reply is produced; the sensor reads it; the loop repeats.

The decay $\rho < 1$ gives the posterior finite memory. Our reimplementaion is standard-library Python against any OpenAI-compatible endpoint; it copies no part of the original codebase and is built only from the public descriptions.

A practical note that matters for reproduction. Some reasoning models return their text in a **reasoning_content** field with an empty **content**. A sensor reading only **content** then silently returns nothing and the loop degrades *without erroring*. Our extractor falls back to **reasoning_content**; anyone reproducing this against a reasoning model must do the same or measure a broken loop.

3 Experiments

All behaviour metrics are ground-truth (word counts, question flags, matches to a known hidden label); where a blind label is needed (E1) it is produced by a separate model that never sees the prefix. Figures below are from a small number of base models and should be read as illustrative in magnitude and indicative in direction.

3.1 E1: structured-placebo ablation

Four arms at matched format: TRUE (production prefix), SCRAMBLED (same posterior with mode labels deranged, dominant and strategy rebuilt so the placebo is internally consistent but wrong), STRATEGY_STRIPPED (true posterior with the strategy line removed; note it still carries the mode names and values, the **Dominant/gate** fields and the “internal stance” framing, so it is *not* a pure numeric control—earlier versions called it NUMBERS_ONLY, which overstated what was stripped), and NULL (no prefix). Under a *weakly* held posterior, TRUE and SCRAMBLED were not separable on the blind-label channel (JSD 0.007 bits against a one-off split-half noise floor of 0.109; no formal equivalence test), although they differ visibly on length (67.5 vs 46.3 words), and STRATEGY_STRIPPED $\approx$ NULL: the imperative *strategy line* carried the visible effect, not the state values. Taken alone this is deflationary—the model follows the instruction whether the state behind it is true or deranged. Scrambling also changes the dominant label, gate text and strategy line together, so it does not isolate a single content factor.

Reproducibility caveat. The deposited E1 summary was produced by an earlier variant of the script with a weakly held posterior; that variant’s exact posterior, generated replies and blind-judge outputs were not retained. The released `e1_placebo.py` clamps the dominant mode at 0.72 and does not reproduce the deposited E1 figures. E1 should therefore be read as illustrative; the strong-state result rests on E2, which is reproducible from the released code.

3.2 E2: clamped factorial

Clamping the posterior to each mode in turn (so the design cannot collapse into a single mode) and toggling gate, monologue and strategy, we ran 30 configurations: 5 modes $\times$ gate $\times$ monologue with the strategy line (20) plus 10 strategy-stripped configurations with the monologue off; each configuration received 2 prompts $\times$ 3 repetitions, i.e. 180 replies in total (an earlier version of this paper wrote “180 cells”; that counted replies, not configurations). The design is a *partial* factorial: the strategy-off $\times$ monologue-on combination was not run, so the strategy $\times$ monologue interaction is not estimable. Note also that “gate off” inserts **gate=full scope** rather than omitting the field, and the monologue condition appends the imperative “Act on that assessment” as well as the monologue text. Under a *strong* clamped posterior:

- the dominant mode **predicts** reply behaviour (Table 1). This is measured with the strategy instruction *on*, so it is largely instruction following by construction; it is a mode-conditioned-style result, not a probabilistic calibration test (no reliability curve, Brier or log-loss), and we no longer call it “calibration”;
- STRATEGY_STRIPPED prefixes now produce long, unshaped replies ($\sim$ 150–200 words) that differ sharply from strategy-bearing prefixes (46–133 words): the state values are *not* universally inert. This **corrects** the E1 reading—inertness was a weak-state artefact;

Clamped mode	mean words	question rate
exploring	51.0	1.00
evaluating	102.2	0.50
overwhelmed	108.0	0.83
asserting	46.0	0.50
disengaging	133.5	0.50

Table 1: E2 mode-conditioned reply style: clamped dominant mode vs. reply behaviour (strategy on, monologue off, gate off; six replies per row). The mode predicts style—exploring is short and always questions; asserting is short and direct; the deliberative modes run long. Because the strategy instruction is on, much of this is instruction following.

- the gate token moves length *substantially* and non-uniformly: disengaging $133.5 \rightarrow 56.0$ words (-58%), asserting $46.0 \rightarrow 78.0$ ($+70\%$), overwhelmed $108.0 \rightarrow 137.0$ ($+27\%$), six replies per mean. An earlier version called this channel “minor”; the data do not support that;
- the monologue shortens the two longest states (disengaging $133.5 \rightarrow 30.2$, overwhelmed $108.0 \rightarrow 76.0$) but *lengthens* asserting ($46.0 \rightarrow 65.5$) despite the asserting strategy asking for one action and stop: the effect is selective, not uniformly state-consistent.

The monologue is the strongest single factor in the factorial where the clamped state is strongest (e2_factorial.json). We stop short of naming it the identified causal channel: the monologue condition also adds text and an imperative, and we ran no matched placebo—equal-length irrelevant text, a monologue generated from a scrambled state, the same monologue without “Act on that assessment”, or monologues replayed across states. Without those, the observed change could reflect extra instruction, semantic priming or generic elaboration rather than processing of a valid self-state. The originating write-ups claim the monologue is causally upstream; our data are consistent with that but do not establish it. It is the leading candidate, not a clean identification.

3.3 E3: hidden-stance game with an honest baseline

Each episode fixes a hidden mode; a simulator writes three short in-mode lines; three predictors are scored against ground truth: random, an honest single-shot LLM reading the whole transcript, and the sensor+filter loop. Over 30 episodes: loop 0.60 (18/30), single-shot 0.60 (18/30), random 0.233 (chance 0.20). Against fixed chance the loop is clearly above it (one-sided binomial $p \approx 1.8 \times 10^{-6}$), though the 95% interval on 0.60 is wide (0.42–0.75). It shows no accuracy advantage over the honest single-shot baseline; equal point estimates do not establish equivalence, and this design does not measure persistence (that would need noisy, conflicting or time-varying evidence), so “the filter buys persistence rather than accuracy” is a hypothesis, not a finding of E3. The task is synthetic (an LLM told the mode writes three lines; a classifier with closely related definitions labels them; simulator, sensor and baseline default to the same base model), and the “loop” here is the per-line classifier plus accumulator, without the prefix-feedback or monologue. The methodological point stands: a bare LLM is *not* at chance on this task, so reporting a chance-level baseline would inflate the apparent gain of the loop.

3.4 E4: closed-loop hysteresis, and the decay confound

Only this experiment probes dynamics. Closing the loop and applying an external push toward *disengaging*, ramped up ($0 \rightarrow 0.9$) then back down ($0.9 \rightarrow 0$) from the same posterior, we measure

push	0.00	0.15	0.30	0.45	0.60	0.75	0.90
up ($\rho=0.97$)	.184	.198	.249	.284	.324	.364	.385
down ($\rho=0.97$)	.382	.405	.417	.421	.419	.422	.401
up ($\rho=1.0$)	.185	.198	.241	.254	.289	.323	.356
down ($\rho=1.0$)	.368	.379	.389	.383	.386	.376	.370

Table 2: E4 disengaging mass along the up/down ramps (one trajectory each). Mean vertical separation 0.126 ($\rho=0.97$) vs. 0.115 ($\rho=1.0$); the separation is essentially unchanged with decay removed.

the disengaging mass at each level (Table 2). The down-ramp sits far above the up-ramp at every level: the mean vertical separation across the seven sampled levels is 0.126 (the result files call this `hysteresis_loop_area`; it is a fixed-grid mean gap, not a geometric area—the trapezoidal area over push 0–0.9 is 0.116). Pushed to disengagement at 0.9 and released to 0, the state remains elevated over this horizon. The up-ramp rises smoothly (max single-step jump 0.05): no large jump is observed between the seven sampled levels, which does not exclude a transition between them. Three design facts bound the reading. The “push” is not an observation shown to the LLM: the code adds $2.5 \times \text{push}$ to the disengaging component of the sensor likelihood, so the estimator is driven numerically and history dependence in an accumulating estimator is expected. Each curve is a single stochastic trajectory (sampling temperature 0.7) without repetitions or confidence intervals. And E4–E6 generate replies from the state prefix alone; they do not call the private monologue, so the dynamics do not exercise the full architecture. So the loop shows substantial **finite-horizon path dependence** but no observed sharp transition.

The decay confound—and why the obvious control is insufficient. The tracker’s decay ($\rho = 0.97 < 1$) builds intrinsic inertia into the posterior, so some stickiness is mechanical rather than emergent. The natural control is a decay-free rerun: with $\rho = 1.0$ the mean separation held at 0.115 (vs. 0.126) —essentially unchanged (Table 2), and the transition remains smooth under both settings. We initially read this as decisive. It is not: an accumulator at $\rho = 1.0$ never forgets, so it is path-dependent *by construction*, and “survives decay-off” cannot distinguish loop-borne memory from accumulator arithmetic. Separating the two requires a null that keeps the accumulator and removes the loop—which is E5b, below.

3.5 E5–E6: what kind of memory?

Three follow-ups decompose the E4 effect. All run at $\rho = 1.0$ on a single base model, with the push held for K turns per level (E4 corresponds to a fast ramp).

E5: ramp-rate sweep. If the path dependence were lag—a viscous state trailing a moving input—slowing the ramp far enough would collapse the separation. Over $K = 1, 3, 6, 12$ turns per level it did not shrink (Table 3). That is the supportable statement: *no collapse was observed for dwell times up to 12 turns per level*. It does not rule out lag. E6 shows relaxation still under way after 100 turns, so $K \leq 12$ is far from the quasi-static limit and the E5 pattern is compatible with very slow convergence. Increasing K also increases the total evidence delivered at each level and, at larger K , reaches the estimator’s concentration cap, so dwell time and evidence dose are confounded in this sweep.

turns per level K	1	3	6	12
closed loop (E5)	0.161	0.210	0.213	0.231
state-decoupled null (E5b)	0.101	0.135	0.142	0.153
loop-attributable excess	0.060	0.075	0.071	0.078

Table 3: Decomposing the path dependence ($\rho = 1.0$; mean vertical separation, one closed-loop trajectory per K). The null—the same accumulator with the loop removed—carries most of the raw separation; the residual is ~ 0.07 at each K and did not collapse over $K \leq 12$. It is an arithmetic difference against an unmatched null, not a measured causal constant.

E5b: state-decoupled null. The decisive control. The identical ramp protocol is run with a likelihood that ignores the state entirely (no LLM in the loop), so any loop area is pure accumulator arithmetic. The null still produces areas of 0.10–0.15 with the same growth in K (Table 3): roughly two thirds of the raw E5 separation is the estimator, confirming that E4’s “survives decay-off” framing overstated what that rerun showed. The closed-loop-minus-null residual is $+0.06$ to $+0.08$ at each K and did not shrink over this range. We report it as an arithmetic residual, not a statistically identified causal effect: the null is deterministic and not matched to the closed loop’s mode and confidence distribution, temporal correlation or API noise, and there is a single closed-loop realisation per K with no uncertainty estimate. A matched control would preserve the real sensor outputs and break only the feedback (replaying or permuting state-to-reply pairings) across many seeds.

E6: two-point basin test. Whether the residual memory amounts to true bistability—two attractors at identical input—is tested by starting the closed loop from opposite dominant modes under a clamped push of 0.45. At $N = 30$ turns the trajectories still held a ~ 0.20 gap, but the lower trajectory was still climbing, so a 30-turn run cannot distinguish two attractors from slow convergence to one (the shipped `e6_bistability.json` verdict, computed at $N = 30$, should be disregarded for exactly this reason). At $N = 100$ the trajectories approach each other (~ 0.60 vs ~ 0.66 disengaging mass), with tail gaps of 0.07–0.08 and the lower trajectory still rising. This is consistent with convergence to a common attractor but the run stops before convergence, and it tests one push value, two initial states, one model and two trajectory pairs. The supportable statement is **no evidence of bistability in this limited test**, not that the system has a single attractor. It does revise E4’s “the state stays in the basin” reading: over a long horizon the state does not stay; the return is slow.

Revised claim. Under a cyclic numerical drive the closed loop shows substantial finite-horizon path dependence, most of which is accumulator arithmetic. A residual of ~ 0.07 (mean separation) remains over an unmatched null and did not collapse for dwell times up to 12 turns per level; its size, mechanism and rate dependence are not established. Initial-condition effects remain visible for over 100 turns with relaxation still under way, and no bistability was observed in the one configuration tested. Every stronger reading tested in this series, including two of our own along the way, failed its control.

4 Discussion

Across these controls the picture is consistent: each tested claim reproduces in a *true-but-softer* form. In small synthetic experiments on one or a few model endpoints, explicit mode-labelled prompt additions changed output length and question use, most strongly when the posterior was confident and with the private monologue as the leading—but not isolated—factor. A classifier-plus-accumulator inferred synthetic modes well above chance without an accuracy advantage over an honest single-shot baseline. Under a cyclic numerical drive the closed loop produced substantial finite-horizon path dependence, most of it accumulator arithmetic, with the size and mechanism of any additional feedback contribution left uncertain; a 100-turn basin test found no bistability and relaxation still in progress. Two of our own intermediate readings (that a decay-free rerun was the decisive confound control, and that a 30-turn basin gap indicated bistability) failed their own follow-up controls and are corrected above; we report them because the failure mode—an accumulator that is path-dependent by construction, and a stability criterion that passes on slow drift—will face anyone reproducing this loop.

None of this speaks to the stronger claims the architecture is often presented with. We measured structural properties of a feedback loop; whether such properties bear on any threshold or phenomenal question is outside what these experiments can address, and we make no such claim. What we can say is narrower and, we think, useful: the loop does several real things, and it does them in a milder and better-controlled form than the original presentation asserts.

Limitations. Figures come from a small number of base models; the directions are the finding, not the magnitudes. The blind labeller in E1 has limited power on the label channel (the word-count and question-rate channels separate more cleanly). E3 uses short (three-line) episodes; longer evidence might let the filter’s persistence convert into an accuracy gain over the single-shot baseline. The E5/E5b/E6 memory decomposition comes from a single base model and one push geometry, from single stochastic trajectories without repetition or confidence intervals; the ~ 0.07 residual is an arithmetic difference against an unmatched null and should not be read as a constant or as an identified causal effect. E4–E6 do not exercise the private monologue. The E2 factorial is partial and its controls are imperfectly named (see E1, E2). The deposited E1 summary is not reproducible from the released E1 script. The implementation is a conceptual partial reimplementaion: it reconstructs the stance tracker, state prefix and monologue from public descriptions but not the second-order interlocutor term or the error-corrector component the originating account describes, and the original code was unavailable for comparison, so architectural fidelity is not verified. Matched placebos for the monologue, a complete factorial, many seeds with uncertainty estimates, genuinely probabilistic calibration tests, and dynamics run well beyond the observed relaxation time are the natural next experiments.

Revision history

v1.1 (September 2026). Revised after an external validity review of v1.0. Corrections of fact: E2 is 30 configurations / 180 replies, not “180 cells”, and is a partial factorial; the `NUMBERS_ONLY` control is renamed `STRATEGY_STRIPPED` to reflect what it removes; the gate channel is substantial (up to ± 60 – 70% in length), not “minor”; the E4/E5 “loop area” is a mean vertical separation across sampled levels. Reproducibility: the deposited E1 summary comes from an unreleased weak-clamp variant. Claims softened to what the designs support: “calibration” becomes mode-conditioned style; the monologue is the leading rather than identified causal channel; E5’s residual is an arithmetic

difference that did not collapse for $K \leq 12$, not rate-independent memory; E6 shows no evidence of bistability rather than a single attractor; E3’s “persistence” is a hypothesis. Title changed accordingly. No data were re-run; all result files are those of v1.0 with corrected notes.

Reproducibility

A clean-room implementation of the loop and all experiments (E1–E6, including the E5b state-decoupled null and the long-horizon E6 rerun), with the raw result files, is released under the MIT license at the project repository. The code uses only the Python standard library and targets any OpenAI-compatible endpoint. The decay-free control is a single environment variable (`SELFMODEL_RHO=1.0`).

Acknowledgements

The self-model loop architecture studied here is due to Lark Laflamme (RavenNest Scientific). This is an independent reconstruction and set of controls; it reproduces none of the original code and is built from the public descriptions. We thank the author for write-ups clear enough to rebuild from.